

Lab Assignment 7

CS 331: Machine Learning

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March 10, 2025. You may work in groups.

This lab will be about linear regression. You need to show your solutions for at least Questions 1 - 3 during the lab session today. In addition to code, please submit a pdf with all your plots and key observations.

1. Generate two datasets each with 100 points where $x \in \mathbb{R}$, $y \in \mathbb{R}$.

- (a) $D1 = \{(x, y)\}$ where $y = ax + b + \text{noise}$
- (b) $D2 = \{(x, y)\}$ where $y = ax^2 + bx + c + \text{noise}$
- (c) $D3 = \{(x, y)\}$ where $y = \exp(x) + \text{noise}$

where a, b, c are inputs and noise comes from a normal distribution $\mathcal{N}(0, 2)$.

2. For each of the three datasets above perform linear regression using a training set of 20 samples, use $(1, x, x^2, \dots, x^d)$ as features where d is to be taken as an input. Use the closed form solution derived in class for linear regression.
 - (a) Compute the least squares error on the training sets.
 - (b) Compute the least squares error on the test sets (consisting of the remaining 80 points).
 - (c) Plot the training data points and the functions learnt for $d \in \{1, 2, 5, 10\}$.
 - (d) Plot the test data points and the functions learnt for $d \in \{1, 2, 5, 10\}$.

Explore which d works best for each dataset for training and test data. Reference for sample plots: [Christopher Bishop, Pattern Recognition, Fig 3.9].

3. Implement linear regression for all the datasets using stochastic gradient descent. Verify if you get the same co-efficient vector as you got using the closed form expression in (2).
4. Perform ridge regression on all the datasets in (2). Use cross validation to choose the value of λ . You may use a training set of 20 points, validation set of 20 points and remaining 60 points as test set. Explore which d works best for each dataset.
5. Explore ridge regression on the wine datasets (white and red wine datasets) using higher dimensional features. Report the best dimensions, best λ using cross validation and least squares error on the test set. Use the following splits: 40% training set + 20% validation set + 40% test. You may download the dataset from the UCI repository <https://archive.ics.uci.edu/dataset/186/wine+quality>